

Wenyin (Knut) Gu

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EDUCATION

- **University of California, Irvine** | 2025 – Present | B.S. in Electrical Engineering (Specialization: DSP)
GPA: 3.64/4.0 | Dean's List (all quarters)
- **Cañada College** | 2022 – 2025 | A.S. in Engineering
GPA: 4.0/4.0 | Valedictorian, *Summa Cum Laude*, Dean's List (all semesters)
- **Stanford University** | Summer 2023 | Fundamentals of Data Science (4-week intensive)
Python, statistics, mathematical optimization, linear algebra, and responsible AI.

RESEARCH

- **Recovering and Combining Partial Multi-Camera Evidence under Wi-Fi Jamming** | EECS 199 Independent Study, advised by Prof. Syed A. Jafar | Jul. 2026
Modeled Wi-Fi jamming as uplink packet erasure and built a 5-camera, 48-step Python simulation (galois, NumPy, OpenCV) splitting each record into coarse and fine Lagrange-coded layers over GF(2003), with T-privacy masking. Layered coding recovered 94% of usable evidence versus 36% uncoded at 3.9× transmission overhead; cross-camera fusion outperformed the best single camera on both accuracy and coverage.
- **“Agentic Affordance: Why LLMs Invite the Intentional Stance” (with Aileen C. Luo)** | 6th International Conference on Philosophy of Mind: Artificial Intelligence (6ICPH) | May 2026
Developed the notion of “agentic affordance” to explain why human–LLM interaction warrants an intentional stance despite a lack of mechanistic interpretability.

PROJECTS

- **Spatial vs. Temporal Edge Detection Simulation** | Mar. 2026
Built Python simulations (NumPy/SciPy) comparing 2D spatial convolution (Sobel operators) with temporal high-pass filtering (difference equations) for edge detection.
Showed that event-based temporal filtering achieves superior signal sparsity and efficiency in motion tracking by mathematically suppressing static backgrounds to zero.
- **Music-Reactive LED System** | Mar. 2026
Designed a custom microphone preamplifier and 3-band active analog crossover (LM358 op-amps) to separate bass, midrange, and treble in hardware.
Programmed a non-blocking Arduino state machine and interactive OLED GUI that turn multi-channel ADC input into three real-time LED visualization modes.

EXPERIENCE & LEADERSHIP

- **Webmaster, IEEE-Eta Kappa Nu (HKN)** | UC Irvine | May. 2026 – Present
Maintain and architect the chapter website, managing updates to event schedules, membership rosters, and technical archives.
- **Communications Subsystem Engineer, Legacy Robotics** | UC Irvine | Apr. 2026 – Present
Audit the rover's 5 GHz video and 900 MHz MAVLink telemetry links for the University Rover Challenge.
Designed a staged upgrade: feedline and antenna fixes worth 6–10 dB of link margin, a GPS- and RSSI-guided auto-tracking base station (ESP32 pan-tilt), and link-aware adaptive video with dual-link failover.
- **Mathematics Tutor** | Cañada College Learning Center | 2022 – 2024
Provided one-on-one and drop-in tutoring, guiding students through problem-solving and analytical reasoning.
- **Community Service** | Shanghai, China | 2007, 2010
Water Supply Team Lead, Shanghai World Expo (2010): led safety inspections of water-supply networks safeguarding drinking water for 73 million visitors over six months; volunteer, Special Olympics World Summer Games (2007).

SKILLS

- **Programming:** Python (NumPy, SciPy, OpenCV, galois), MATLAB, C++, Verilog.
- **Hardware & EDA:** Analog circuit design (op-amps, active filters, BJT/MOSFET), Cadence Virtuoso, SolidWorks, Onshape, Arduino/ESP32 integration (ADC, I2C, OLED).
- **Communications:** RF link budgeting and antenna selection, OFDM, erasure and Lagrange coded computing, Mission Planner.

HONORS AND AWARDS

- IEEE-Eta Kappa Nu (2026) · Phi Theta Kappa (2023)
- Dean's List (2022 – Present)
- Valedictorian, Cañada College (2025)